

Decoding Deep Sampled Brains: High-Accuracy Binary Emotion Classification from Audiovisual Stimuli

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RESULTS

Emotion categories were decoded versus neutral with **high within-subject accuracy across repeated fMRI sessions**. Transformed weight maps showed distributed and **individually specific neural activation profiles** that differed both between emotions and between subjects.

TAKE-HOME MESSAGE

Emotion recognition is supported by **stable but individually specific** brain-wide patterns, highlighting the **value of deep-sampling single individuals**.

INTRODUCTION

- Group averages obscure individual responses (Fischer et al., 2024)
- Functional brain organization is individual specific (Gordon et al., 2017)
- Functional brain networks are more stable within individuals than across tasks (Gratton et al., 2018)
- Emotion categories are not localized to discrete brain regions (Lindquist et al., 2012; Barrett, 2017)

Research questions

- Is there a stable neural signature within subject across emotion categories?
- Are the neural signatures
 - separable between conditions within-subjects?
 - separable between subjects within the same condition?
- Which brain regions are involved?

Two Individuals

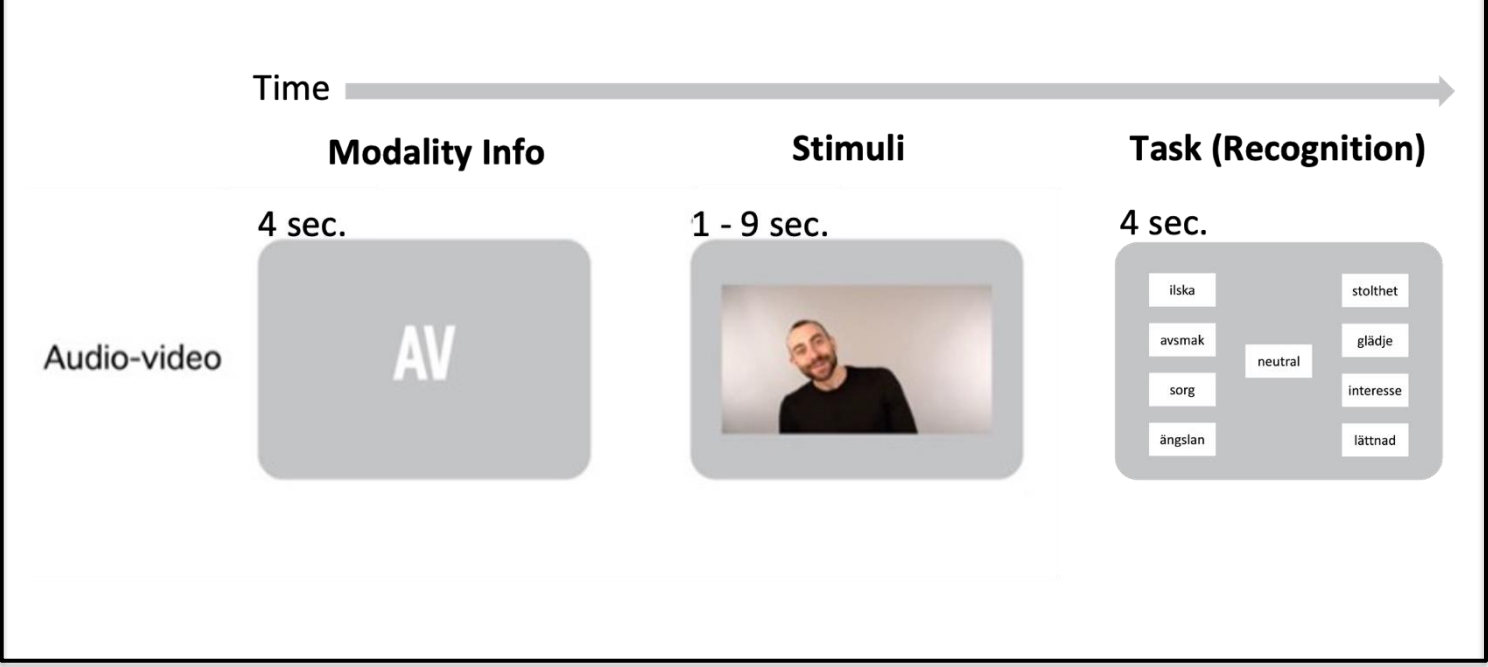

Subject A
age: 25
female


Subject B
age: 25
male

- 18 common fMRI scanning sessions each
- identical stimuli sets per session

EXPERIMENT

Paradigm



Stimuli

- clips of actors performing emotion expressions
- 9 emotion categories
 - anger, anxiety, disgust, sadness
 - neutral
 - happiness, pride, interest, relief
- audiovisual stimuli
- 12 unique stimuli sets
 - 1 set per session
 - sets are repeated after 12th session

ANALYSIS

GLM

- 1st level beta maps
- derived for each condition

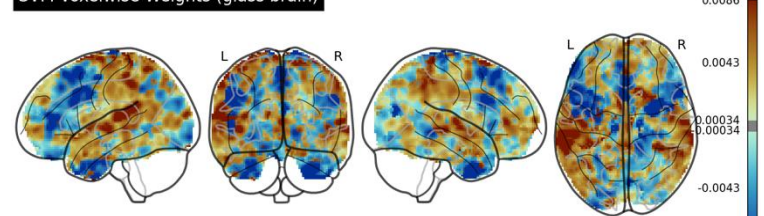
SVM

- condition beta maps as input
- binary classification (Schrouff et al., 2013)
 - each emotion vs. neutral
 - for 18 sessions
- Leave-One-Session-Out CV
 - 18 folds
- 1000 permutations
- outcomes:
 - accuracy
 - weight maps

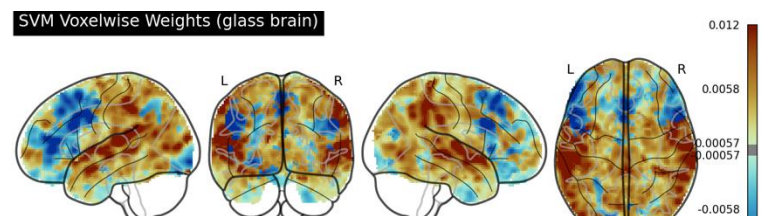
Transform

- transform SVM weight maps into physiologically interpretable maps
- Haufe-Transform (Haufe et al., 2014)
$$A = \Sigma_x W \Sigma_s^{-1}$$
- covariance between beta values and SVM weight coefficients

SVM weight map example:

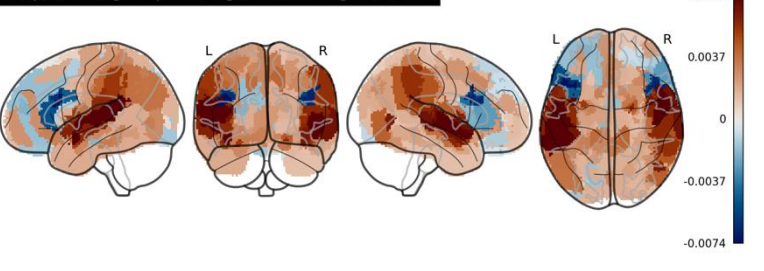


Transformed map example:



Comparison

Thresholded Maps Comparison:

- Jülich brain atlas parcellation of transformed maps
- signed mean per parcellation
- threshold maps to show the top 20 parcels in terms of absolute signed mean

Spearman Rank Correlation:

- Pairwise Spearman Rank Correlation
 - within-subject
 - between-subject

Network Profiles:

- in ROIs associated with Emotion Recognition (Dricu & Fröhlich, 2020)

RESULTS

SVM Emotion Classification Accuracy (18-fold CV)						
Participant	Emotion	Accuracy	p	N Folds	Fold SD	Class 1
Subject A	Anger	1.00	< .001	18	0.00	1.00
	Anxiety	0.81	< .001	18	0.30	0.78
	Disgust	0.94	< .001	18	0.16	1.00
	Happiness	0.89	< .001	18	0.21	0.83
	Interest	0.92	< .001	18	0.19	0.83
	Pride	0.67	= .040	18	0.33	0.61
	Relief	0.75	= .009	18	0.25	0.67
	Sadness	0.81	= .003	18	0.24	0.67
	Sadness	0.81	= .003	18	0.24	0.67
Subject B	Anger	0.89	< .001	18	0.21	0.78
	Anxiety	0.75	= .008	18	0.25	0.67
	Disgust	0.83	< .001	18	0.24	0.94
	Happiness	0.92	< .001	18	0.25	0.94
	Interest	0.92	< .001	18	0.19	0.83
	Pride	0.94	< .001	18	0.16	0.94
	Relief	0.89	< .001	18	0.27	0.83
	Sadness	0.83	< .001	18	0.24	0.83
	Sadness	0.83	< .001	18	0.24	0.83

Between-subject Spearman p: Subject A vs Subject B												
	anger	anxiety	disgust	sadness	happiness	pride	interest	relief				
anger	0.36***	0.02	0.04	0.21***	0.31***	0.11*	0.11	0.24***				
anxiety	0.41***	0.08	0.11*	0.24***	0.40***	0.11	0.12*	0.25***				
disgust	0.29***	0.03	0.11*	0.38***	0.36***	0.12*	0.16**	0.25***				
sadness	0.04	-0.08	0.06	0.16**	0.07	0.05	-0.08	0.03				
happiness	0.15**	-0.01	0.12*	0.35***	0.32***	0.18**	0.11	0.27***				
pride	0.45***	0.33***	0.30***	0.21***	0.30***	0.26***	0.40***	0.35***				
interest	0.45***	0.15*	0.14*	0.24***	0.42***	0.17**	0.23***	0.30***				
relief	0.27***	0.14*	0.06	0.01	0.22***	0.10	0.14*	0.21***				

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Subject A

Subject B

Top 20 ROIs - Transformed Maps

Emotion Recognition ROI Profiles



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